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WT-1200I

Hand-held Recombination Lifetime Measurement Tool

User Manual



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Disclaimer: The product image on the front page is for illustration purposes only. The actual product may vary.

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1. Introduction

The WT-1200I is a single-point non-contact minority carrier lifetime measuring device capable of monitoring defects and contamination both in the bulk and in the surface region of silicon ingots. It is primarily optimized for the investigation and quality control of ingots, but can also be used with wafers and solar cells.

1.1 Product features

- Fully automatic operation and data evaluation
- Measurement selectable at any position on the sample
- Measurement of mono-crystalline material
- Ingots can be measured
- 1:1 correlation with the standard Semilab WT-2000PI Ingot Tester

1.2 Specifications

Table 1-1: Indoor environment

Operating temperature	15-35°C
Relative air humidity	10-80%
Maximum height above sea level during operation	2000 m
Clean room	Dust-free environment equivalent to Class ISO 9
Vibration sensitivity	VC-A

For detailed specifications of other system components, refer to their respective manuals. Specifications in next paragraphs concern the main unit only.

Table 1-2: Power line supply

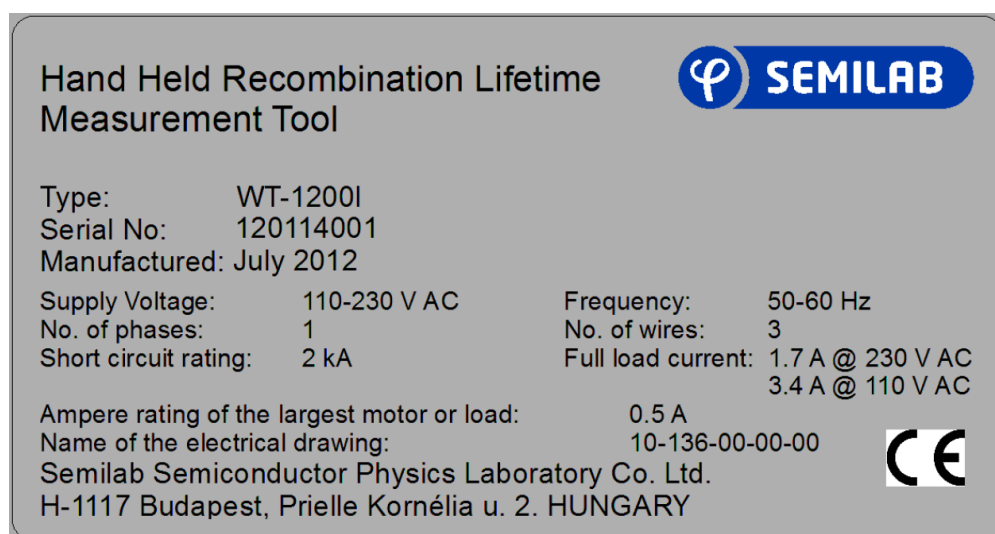
Line type	Single-phase, phase-neutral-protective ground system
Line frequency:	50-60 Hz
Number of phases	1
Number of wires	3
Max current	1.3A @ 230VAC / 2.7A @ 110VAC
Input voltage	115 VAC or 230 VAC $\pm 10\%$
Power consumption	300 VA maximum
Power line connection	1.8 m long, 3-wire 3 x 0.75 mm ² power cable
Class of hazardous electrical shocking	class I, protective conductor with green/yellow color
Overvoltage group	II, according to IEC 664

Table 1-3: Overall size of equipment

Computer	436mm x 522mm x 178mm
Laser unit	485mm x 480mm x 140mm
Laser head	65mm x 48mm x 135mm

Table 1-4: Weight of equipment

Computer	17.1kg
Laser unit	5.7kg
Laser head	0.3kg

**Figure 1-1: CE dataplate**

2. Safety Precautions

The WT-1200I is designed to be completely safe when operated and maintained by properly trained personnel adhering to the procedures specified by Semilab. Semilab assumes no responsibility or liability for personal injury or damage to the system originating from user-adopted operating procedures or safety procedures. The user is responsible for operating the WT-1200I system in a safe manner.

In this chapter, Semilab provides safety information regarding the WT-1200I system design and its intended use. Read this chapter thoroughly before operating or servicing the system.

2.1 Conventions

This manual includes three types of statements that require the user's attention. These are enclosed within a border for emphasis.

✓ **NOTE:** Note statements highlight or clarify information.

! **CAUTION:** Caution statements identify conditions or practices that may result in damage to the system or to the wafers being processed.

 **WARNING:** Warning statements identify conditions or practices that may endanger personnel, resulting in injury or loss of life.

This manual uses several conventions for software interface instructions. These are listed below:

Table 2-1: Conventions

Operator Interface	Convention
Software menu listings	<i>bold italic text</i>
Buttons and areas of the window or screen that are selected by clicking with the mouse	bold text
Names of windows and screens	<i>italic text</i>
Names of areas within windows and screens	<i>italic text</i>
Keyboard entries	<text in angle brackets>
Hardware buttons	<u>UPPERCASE BOLD UNDERLINED TEXT</u>
Unique names and terms	[text in square brackets]

2.2 Safety symbols

The symbols indicated in Table 2-2 appear within this manual and on safety labels at various locations on the WT-1200I system. These symbols are used to alert the user of potential safety hazards.

Table 2-2: Safety symbols

	Ground (earth) terminal
	Electric shock hazard
	Laser radiation hazard
	Caution, risk of danger

2.3 Laser safety

The WT-1200I contains a high-power continuous-wave infrared laser source; its radiation is invisible to the human eye.

⚠ WARNING: Do not bring the bottom of the measuring head close to the eyes or any part of the body. Do not make direct eye contact with the light source.

The WT-1200I equipment is a Class 1M laser product, classified according to EN 60825-1 standard. The laser beam is highly divergent, but it can be accessed by incorrect handling. It is safe for the naked eye if the distance from the eye is at least 10 cm, potentially hazardous when optical instruments are used or if it is brought close to the eye. The Laser information label is printed on the measurement head, and the aperture is also indicated by a label as seen in Figure 2-1.



Figure 2-1: Warning labels on WT-1200I measurement head and aperture

The laser diode of the PCD unit is built in the laser unit box. The invisible laser radiation is emitted from the measurement head down perpendicular to the sample, which is placed below the measuring head. During normal operation there is a low risk of exposure to laser radiation.

Measuring head as well as laser diodes do not require maintenance. In case of removing any part of the protective cover from the industrial PC or dismounting the head unit, scattered reflected invisible laser radiation might appear. Normally it is not harmful, but should be avoided viewing directly to the fiber connection, the bottom of the measuring head or using viewing optics to inspect any part of the equipment around the connector and the under operation.

2.4 Risk assessment

Applied standards

- *ISO 14121-1:2007, Safety of machinery — Risk Assessment — Part 1: Principles*
- *ISO/TR 14121-2:2007, Safety of machinery — Risk Assessment — Part 2: Practical guidance and examples of methods*

Introduction

The WT-1200I system is a table-top tool that can measure the minority carrier lifetime in silicon ingots. It is a useful tool for incoming ingot inspection, quality control and process monitoring in the ingot manufacturing processes.

The WT-1200I can be equipped with the following measurement options:

- e-PCD for determination of minority carrier lifetime

Limits of the machinery

a.) assembly and commissioning (only by Semilab's engineer)

- connection the parts of the system together
- connection to the electric power supply
- checking the functioning of all controls and the proper operations of the machine

b.) operation (environmental limits)

- ambient temperature between 15°C and 35°C, varying <1°C per hour
- relative humidity between 10 % and 80 %
- maximum attitude above sea level during operating: 2000 m

c.) Maintenance and service should only be performed by trained and qualified service personnel.

Use limits

a.) intended use

- the equipment is intended for measuring different shape/size of silicon ingots.
- the equipment is intended to be used by trained persons.
- the equipment is intended to be maintained by Semilab trained service engineer.

b.) reasonably foreseeable misuse

- processing non-conform materials other than those intended by the manufacturer
- using the machine with home-made tools
- replacing components or spare parts with ones other than those specified
- use of the machine by not trained persons
- using by out of specification facilities (provided with the machine)

Space limits

The machine is intended for use in an indoor environment.

For installation and use a flat area of at least 1x2m is required (free of obstacles).

Time limits

Warming up time (30 minutes) directly switching on the complex system before measuring. During the period July 12, 2012 to July 18, 2012 a detailed information risk assessment was performed on the WT-1200I system. The assessments can be reviewed in the following pages.

Description of the method used during risk estimations

Before the risk is estimated using the risk graph, the associated hazard, hazardous situation, hazardous event and possible harm is described in accordance with the above mentioned standard *ISO 14121-1:2007*. A risk index is then calculated using the Risk Graph given in Figure 1., based on the following four parameters:

- Severity of harm: S
 1. S1 - slight injury (usually reversible)
 2. S2 - serious injury (usually irreversible, including fatality)
- Frequency and/or duration of exposure to hazard: F
 1. F1 - twice or less per work shift or less than 15 min cumulated exposure per work shift
 2. F2 - more than twice per work shift or more than 15 min cumulated exposure per work shift
- Probability of occurrence of the hazardous event: O
 1. O1 - mature technology, proven and recognized in safety application
 2. O2 - technical failure observed in the two last year
 3. O3 - technical failure regularly observed (every six months or less)
- Possibility of avoidance or reduction of harm: A
 1. A1 - possible under some conditions
 2. A2 - impossible

After risk reduction, a Risk Index value of 1 or 2 should be reached. This means there is no need for further actions in the context of risk reduction.

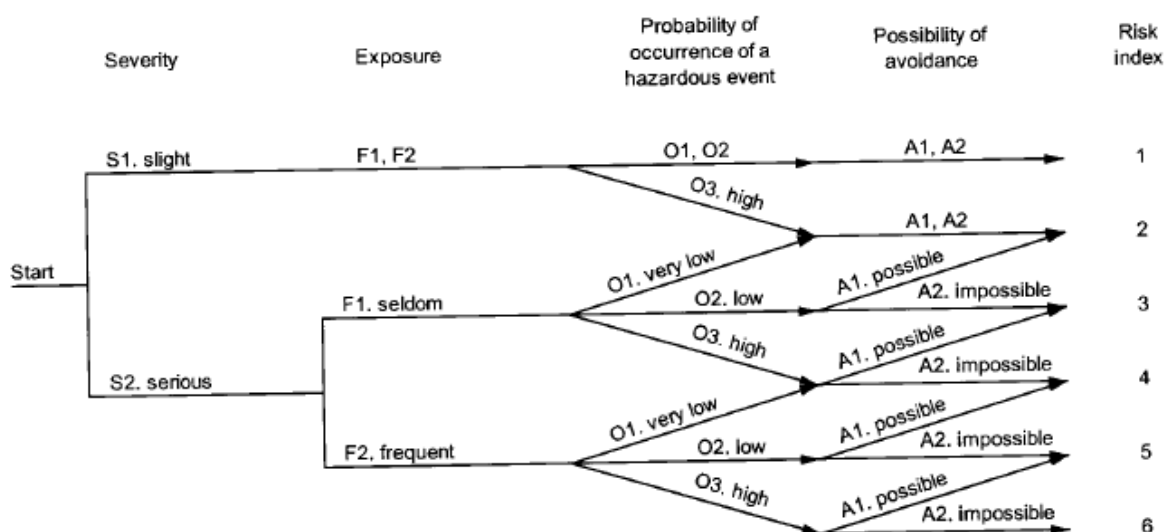


Figure 2-2: Risk graph

Risk assessment (hazard identification)							
Machine		WT-1200I system		Analyst		Zsákai Gyula	
Sources		Manuals, designers and engineers		Current version		3.0	
Extent		Use phase: service, installation and operation		Date		July 18, 2012	
Method		Checklists: ISO 14121-1:2007, Annex A		Page		1.	
Ref. no.	Life cycle	Task	Hazard zone	Hazard	Accident scenario		Ref. no.
					Hazardous situation	Hazardous event	
1	Use phase: Operation	under measuring	machine surrounding	laser radiation	looking into laser beam	laser radiation	1
2	Use phase: Operation	under measuring	machine surrounding	radio frequency radiation	being next to the equipment	radio frequency radiation	2
3	Use phase: Service	service phase	laser unit box	laser radiation	looking into laser beam	laser radiation	3
4	Use phase: Operation	under measuring	optical fiber	laser radiation	looking into laser beam	laser radiation	4
5	Use phase: Installation	unpacking	machine surrounding	injuring	lifting the tool	injuring of the back	5

Risk assessment (risk estimation and risk evaluation) and risk reduction														
Machine			WT-1200I system				Analyst			Zsákai Gyula				
Sources			Manuals, designers and engineers				Current version			3.0				
Extent			Use phase: service, installation and operation				Date			July 18, 2012				
Method			Risk graph (ISO/TR 14121-2 A4)				Page			1.				
Ref. no.	Risk estimation (Initial risk)					Risk reduction	Risk estimation (Risk after reduction)					Further risk reduction required	Ref. no.	
	S	F	O	A	RI		S	F	O	A	RI			
1	1	1	3	1	2	Laser beam is highly divergent; the laser output energy per pulse is under the AEL limit belongs to the Class1M. Warning labels are stuck on the system.	1	1	3	1	2	No	1	
2	1	1	3	1	2	The radiated power is low.	1	1	3	1	2	No	2	
3	2	1	3	1	3	Additional cover on the laser unit; warning label on the laser unit.	2	1	1	1	2	No	3	
4	2	1	1	1	2	Using damage-resistant optical fiber and additional metal cover.	2	1	1	1	2	No	4	
5	1	1	1	1	1	The weight of the tool is small enough to avoid injuries. Moving the system can be done manually.	1	1	1	1	1	No	5	

3. Theory of Operation

The technique used by the WT-1200I involves optical excitation and signal detection by the microwave system.

3.1 Laser excitation

The pulse of an infrared semiconductor laser generates free electron-hole pairs under the illuminated area on the sample (Figure 3-1). Since the penetration depth of the 980 nm wavelength light is about 100 μm in silicon, the free carriers are generated close to the front surface of the sample.

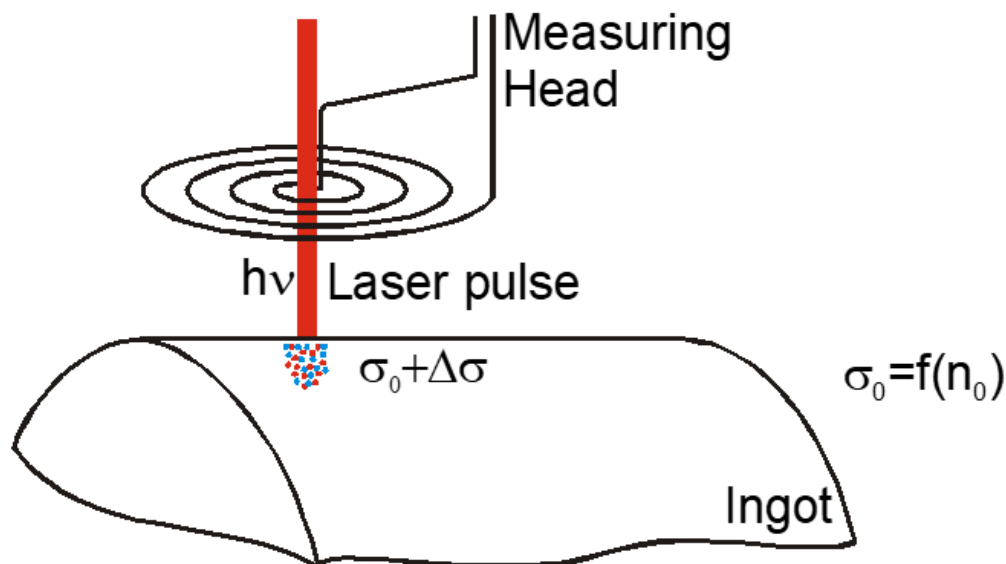


Figure 3-1: Laser excitation

3.2 Detection

Since the free electrons and holes recombine, their concentration and thereby the conductivity of the sample decreases after the excitation. The decaying conductivity can be monitored by producing eddy current and detecting the dissipated energy, because it depends on the conductivity of the sample (Figure 3-2). It is measured as a function of time.

When an AC current flows in a coil in close proximity to a conducting material the magnetic field of the coil will induce circulating (eddy) currents in the sample. The conductivity of a material has a very direct effect on the eddy current flow: the greater the conductivity of a material, the greater the flow of eddy currents on the sample. The eddy current measurement is actually the measurement of the electrical loss in the material.

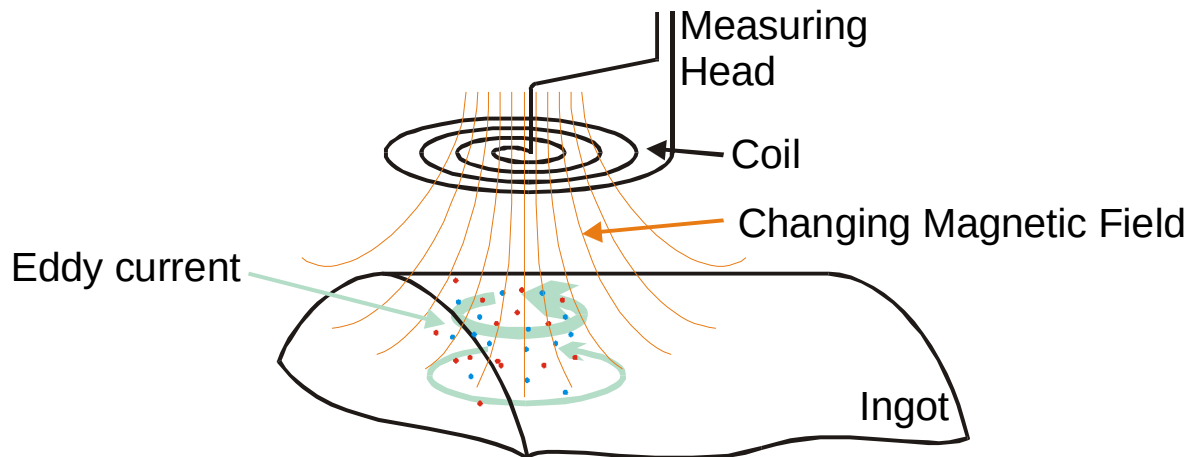


Figure 3-2: Detection by eddy current

The output signal of the system is proportional to the power, which is necessary to keep the amplitude of the oscillation in the coil constant. This signal has its maximum at a given resistivity and decreases towards the very small (e.g. metals) and very high (dielectric) resistivity materials. The eddy-PCD works in the resistivity range that is higher than the resistivity of the maximum signal. Therefore, samples whose resistivity exceeds $0.5 \Omega\text{cm}$ may be measured with the e-PCD.

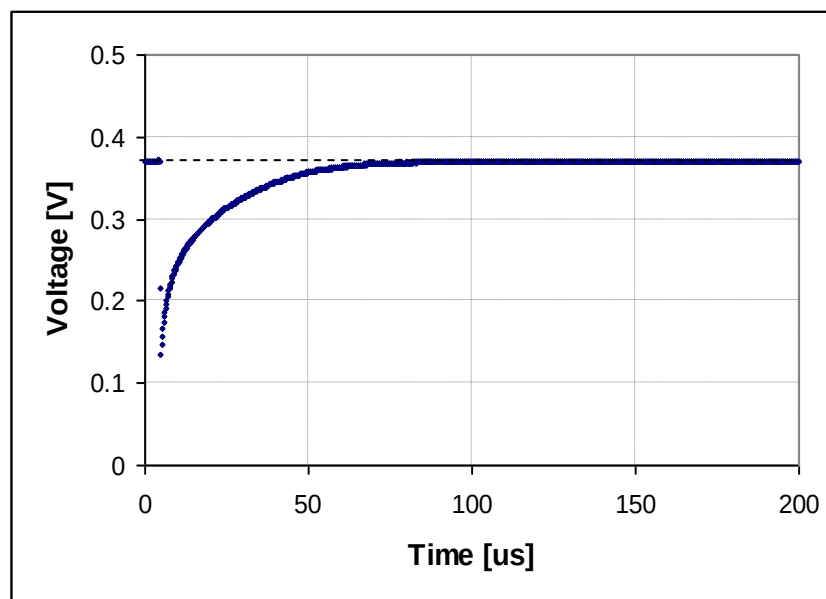


Figure 3-3: Transient, measured on a sample. As the carriers return to equilibrium exponentially, the signal changes accordingly.

The signal is lower right after illumination than in the equilibrium, the signal is increasing during recombination, as it is indicated on Figure 3-3.

3.3 Evaluation

The measured dissipation decay is fitted with a modified exponential curve and the obtained time constant is recorded as the measured, effective lifetime in the given position of the sample.

The fitted curve is the following:

$$U(t) = A \frac{\exp\left(-\frac{t}{\tau}\right)}{\left(1 + \frac{t}{t_0}\right)^n} + O, \quad (3.1)$$

Where A is the amplitude, τ is the lifetime, O is the offset, t_0 and n are characteristic to the diffusion of the carriers from the generation place at the surface, to the bulk. All τ , A , t_0 , n and O are fitted, τ is evaluated as lifetime.

3.4 Bulk and surface recombination

The redistribution (homogenization) of the free carriers takes place during and immediately after the excitation. After this relatively fast event there are two main parallel processes until the thermal equilibrium is reached (Figure 3-4):

- bulk recombination
- carrier diffusion to the sample surface and then surface recombination

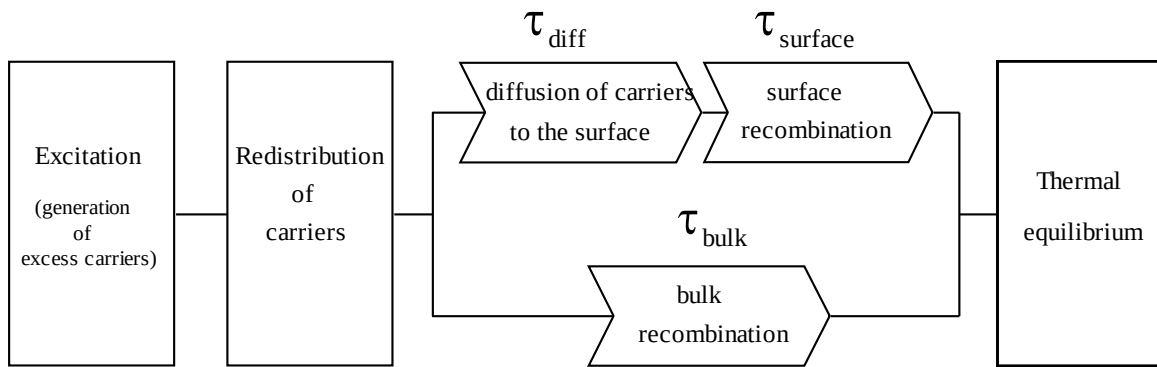


Figure 3-4: Block diagram of the sequence of processes during the μ -PCD measurement

The measured lifetime characteristic of the whole recombination process as it is described with the following equations:

$$\frac{1}{\tau_{meas}} = \frac{1}{\tau_{bulk}} + \frac{1}{\tau_{diff} + \tau_{surf}}$$

$$\tau_{diff} = \frac{d^2}{\pi^2 \cdot D_{n,p}}$$

$$\tau_{surf} = \frac{d}{2 \cdot S}$$

Where:

- τ_{bulk} is the bulk recombination lifetime that is characteristic of material quality.
- τ_{diff} is the characteristic time necessary for carrier diffusion to one of the surfaces of the wafer from the place of excitation. It is determined by the wafer thickness and the diffusion constant of minority carriers.

- τ_{surf} is the characteristic time of surface recombination, determined by surface recombination velocity. It is short for polished bare wafers ($S \approx 50000 \text{ cm s}^{-1}$) and long for passivated wafers ($S < 10 \text{ cm s}^{-1}$).
- τ_{meas} is the measured lifetime.
- d is wafer thickness.
- D_n and D_p are diffusion constants of the minority carriers, for electrons and holes.
- S is the surface recombination velocity.

As demonstrated in the equations above, the sample surface may have a large influence on the lifetime because of the surface recombination process. Figure 3-5 shows theoretical data of the basic equation calculated for three different surface recombination velocities.

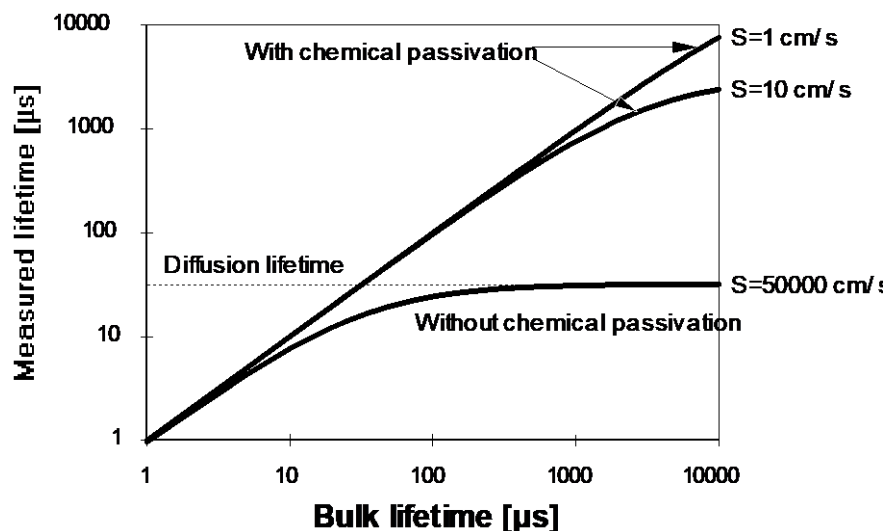


Figure 3-5: Theoretical correlation curves of the measured lifetime versus bulk lifetime

The two steps of the surface recombination process—the free carrier diffusion to the surface and the recombination at the surface—are described with two independent parameters (τ_{diff} and τ_{surf}). If the surface recombination velocity is very high, τ_{surf} is very small (typically on bare surfaces), the carrier diffusion is the limiting process, i.e. τ_{diff} describes the entire surface recombination process. The longer the bulk lifetime, to the greater extent the diffusion lifetime determines the measured lifetime. Since the aim of the measurement is to determine the contamination dependent bulk lifetime, the surface recombination process should be suppressed i.e. the sample surface should be passivated.

There are two commonly used methods of surface passivation: thermal oxidation and chemical passivation. Depending on the efficiency of the passivation, bulk lifetimes much longer than the diffusion lifetime can be measured. Although the measured lifetime depends on both the bulk and the surface recombination processes, in special cases it refers to either the bulk or the diffusion lifetime. Approximately the bulk lifetime is measured if the measured lifetime is much shorter than the diffusion lifetime (high contamination level) independently of the surface recombination velocity. The bulk lifetime is measured in the case of passivated surface as well. The diffusion lifetime is measured if the surface recombination velocity is high (bare sample) and the bulk lifetime is much longer than the diffusion one (low contamination level). In this case the obtained lifetime map is very homogeneous since the diffusion lifetime depends only on the sample thickness and the diffusion constant of the minority carriers which parameters typically do not change a lot in a sample.

3.5 Lifetime measurement with the WT-1200I

The WT-1200I is a fully computer-controlled equipment. The measured transients are recorded by the transient recorder card.

Semilab's eddy sensor design provides unsurpassed sensitivity and measurement speed, and allows the introduction of new evaluation principles. The entire eddy system is integrated into the WT-1200I measuring head.

3.6 Automatic parameter setting in the e-PCD measurement

The automatic setting of the measuring parameters (*AutoSetting*) needs to find an optimal setting taking into consideration the above problems. The best possible "wide-band" evaluation is necessary, that is in a concrete Time Window the lifetime should be evaluated even if it changes orders of magnitude, supposing that the Voltage Range is set correctly.

The AutoSetting procedure works in serial mode i.e. the individual parameters are set in turn.

The parameters are set to their initial value:

- Afterwards the Voltage Range is decreased until the amplitude of the transient (or its recorded first part) is optimal for the A/D conversion, and at the same time the extreme values of the transient are also far from the scope borders. (AutoRange)
- An approximately optimal Time Window can be found which results in valid lifetime data within the Time Window. Especially in case of strongly distorted transients, it can occur that the resulting Time Window is longer than the optimal one. After the Time Window setting the number of averaging is corrected to an approximate optimum value. (AutoTimeWindow)
- The above steps are repeated to find the optimal parameter settings.

4. Installation

The WT-1200I system is a computer-controlled equipment which is suitable for recording, storing and displaying minority carrier lifetime of silicon ingots.

4.1 System configuration

Basic units:

- Laser unit box with measuring head
- Industrial PC (Windows operation system)
- Peripherals (keyboard, mouse, LCD monitor)
- μ -PCD cable
- Serial cable
- Power cable
- Main cord for industrial PC
- Main cord for laser control box
- Main cord for monitor

 **WARNING: Do not use main cords other than those provided for the system by Semilab. Use the main cords in accordance with how they are labelled.**

4.2 Unpacking, inspecting and preparing system components for installation

4.2.1 WT-1200I main unit

1. Unpack and remove the WT-1200I measuring head with the laser unit box, the industrial computer, the cables and other units like: the monitor, the keyboard, the mouse etc. from the crates.
2. Check the crate contents according to the packing list.
3. Inspect the system if there is any damage observable. Report damage to shipper immediately.
4. Make sure that the exterior of the instrument was not damaged. Any deformation on the exterior of the instrument can influence its operation.

 **NOTE: If any part of the instrument has become loose during shipment, it can cause inaccuracies in the measurement results. Tighten them before use.**

4.2.2 Other components of the system

Unpack all other components of the system and refer to their respective manuals for instructions (included in the package).

4.3 Installation

4.3.1 System power and interconnections

 **WARNING:** All electrical connections of the system should be performed with the power switched off and the line cable unplugged.

The following list provides a short description of the cables and their interconnection. For socket numbering, refer to **Error! Reference source not found.**6 below the cable descriptions.

- *Mains cable.* Two mains power cables are supplied with the tool. Connect them to the power sockets of the industrial PC and the laser unit box.
- *μ -PCD cable.* This cable connects the WT-1200I laser unit box and the transient recorder in the industrial PC. One end of this cable should be plugged into socket no. **1.** (Figure 4-4) on the industrial computer and the other into the socket **2** of the laser unit box.



Figure 4-1: μ -PCD cable

- *Serial cable.* This cable makes connection between WT-1200I laser unit box and the serial port in the industrial PC. One end of this cable should be plugged into socket no. **3.** (Figure 4-4) on the industrial computer and the other into the socket **4** of the laser unit box.

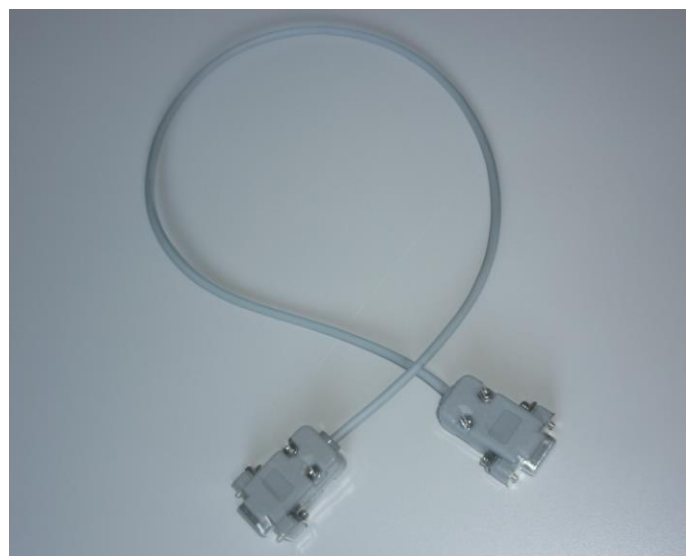


Figure 4-2: Serial cable

- *Power cable.* This cable makes power connection between WT-1200I laser unit box and the industrial PC. One end of this cable should be plugged into socket no. **5**. (Figure 4-4) on the industrial computer and the other into the socket **6** of the laser unit box.



Figure 4-3: Power cable

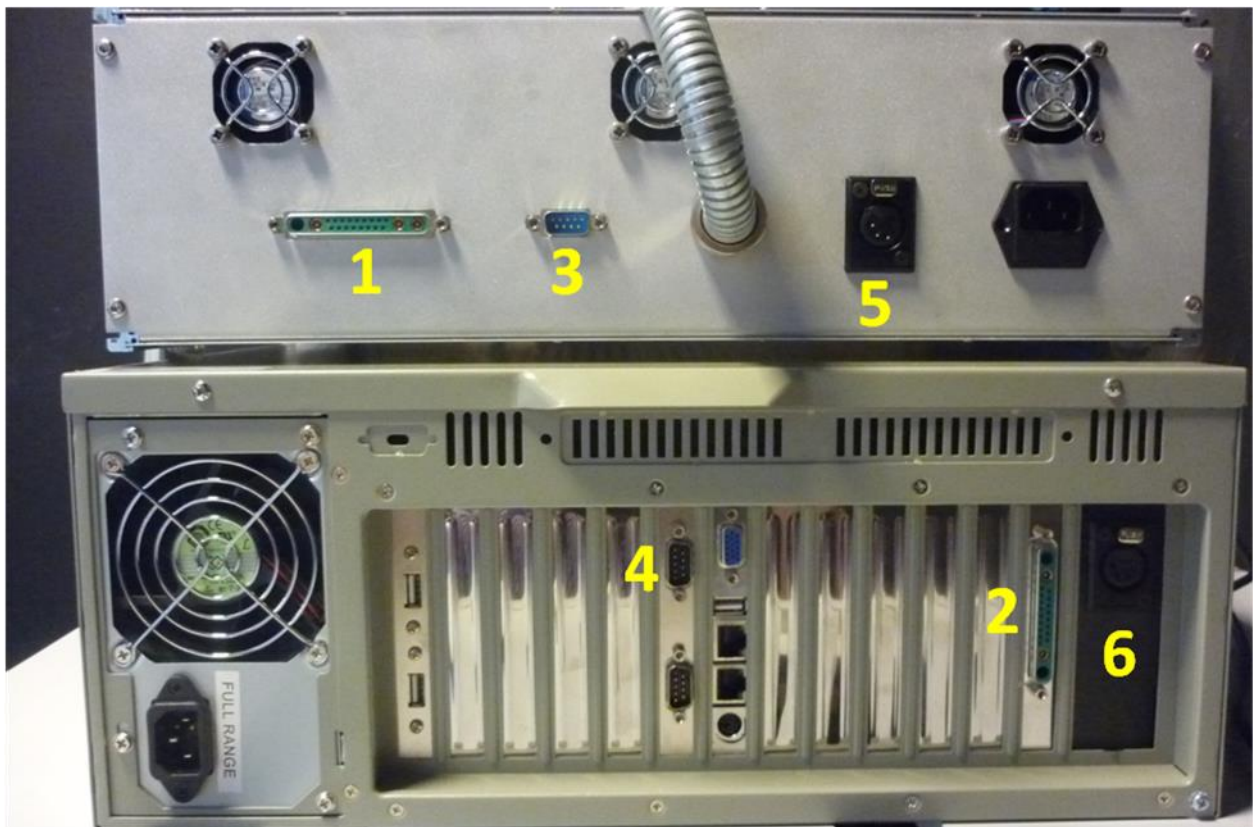


Figure 4-4: Industrial PC and laser unit box back panel

4.4 Software installation

The software is installed in *C:\uPCD* on the computer shipped with the WT-1200I system. The WinDriver program should also be installed for correct operation. The measuring program can be started with the μ -PCD icon on the desktop.

- ✓ **NOTE:** *If the WT-1200I system is purchased with the standard configuration and shipped with an industrial PC, the software comes pre-installed on the PC. Reinstalling the software is not recommended, because this removes the system configuration files.*
- ✓ **NOTE:** *An install CD for the software is supplied in case reinstallation becomes necessary.*

Since the system is controlled by a PC running the MS Windows XP or Windows 7 operating system, a special driver must be installed to enable communication between the hardware and software. This control driver is called “WinDriver”. If the main software cannot detect Windriver installed and properly configured it will pop up a corresponding error message and will start in demo mode (no real measurement will be possible).

To reinstall on the computer, insert the Inline Install CD in the proper drive, and follow the procedure described in the Windriver install for Win7 or the Windriver install for Windows XP.

5. Operation

The bottom of the sensor must fit to the sample. Therefore, the entire measurement head sits on the surface of the sample during the measurement.



Figure 5-1: Sample measurement

The black plastic part of the sensor touches the sample.

Before the e-PCD measurements, do not forget to use the **AutoSetting** procedure as follows:

1. Put the head on the surface of the sample, and hold it steadily.
2. Switch to the *Recorder* window.
3. Click **Autosetting**.
4. Do not move the head until the plot changes. Then the autosetting is finished.

In general, autosetting is required before all measurements and for all samples. If the samples are from the same production line and have very similar properties, the same settings may be suitable for them. In these cases, autosetting can be skipped after the first sample to speed up the measurement.

Whenever the head is placed on a surface, a measurement is performed if the optical sensor is turned on (default).

New statistics can be created in the *Runtime* window; the file name of the statistics must be provided in advance. A new measurement is performed by clicking **New**. The value is automatically added to the statistics.

The measured values appear in a square at the left of the value indicator, at the bottom of the window. The color of the square indicates the relation of the measured value to the classification.

5.1 μ -PCD operating software requirements

For the software to operate as intended, the following requirements must be met:

Software requirements:

- MS Windows XP or Windows 7.

Hardware requirements:

- Pentium compatible computer running the Microsoft Windows operating system
- Optional windows compatible printer

5.2 User Interface and basic operation

Launching the μ -PCD software displays the main window.

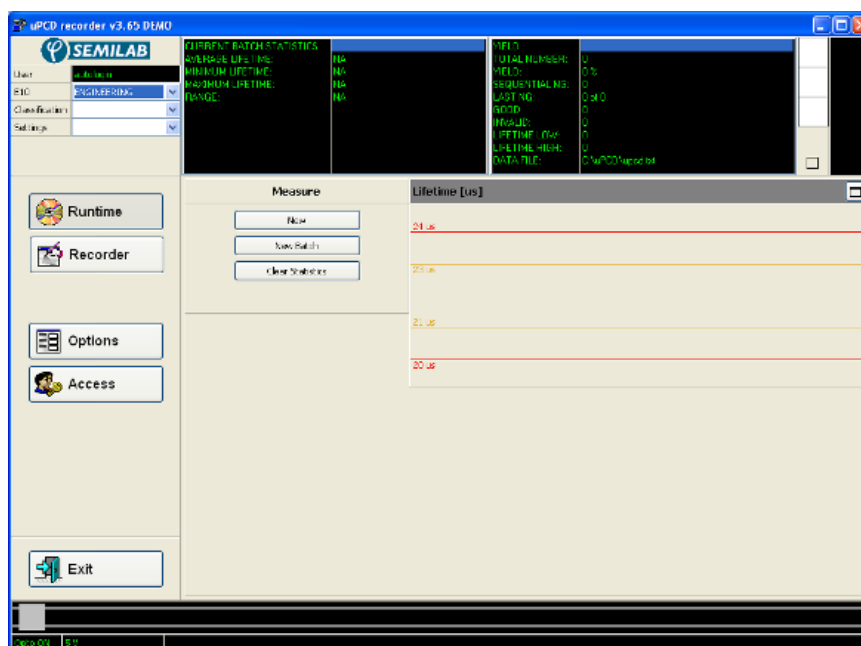


Figure 5-2: Main window/Runtime

The window consists of four main parts:

- *General information and Statistics area* in the top left corner;
- *Mode selection*: left side, below *General information*;
- *Working area*: Middle of the window;
- *Value indicator*: Bottom of the window.

5.3 Working area

The type of the working area of the μ -PCD software can be selected by the four buttons of mode selection: **Runtime**, **Recorder**, **Options** and **Access**.

5.3.1 Runtime

The measured values are indicated on the plot on the right side. Red lines denote the stop limits, orange lines denote the warning limits.

If the measured lifetime is within the warning limits, it is regarded as good and marked as a green dot on the plot and a green square in the value indicator. If the value is between the upper (or lower) warning limit and the upper (or lower) stop limit, an orange dot is put on the plot, and an orange square appears in the value indicator. If the value is outside the stop limits, all these colors are red. The limits can be set in options; see section 5.3.3.1 for further information.

There are three buttons in the window.

- **New** performs a new measurement, the measured value is added to the statistics.
- **New Batch** starts a new set of statistics, but the values are recorded to the same statistics file.
- **Clear Statistics** closes the current statistics file, clears all data from the statistics, and asks for a new statistics file name.

5.3.2 Recorder window

The measuring parameters and the plot window—which displays the transient in the *Transient Recorder* window—are found here.

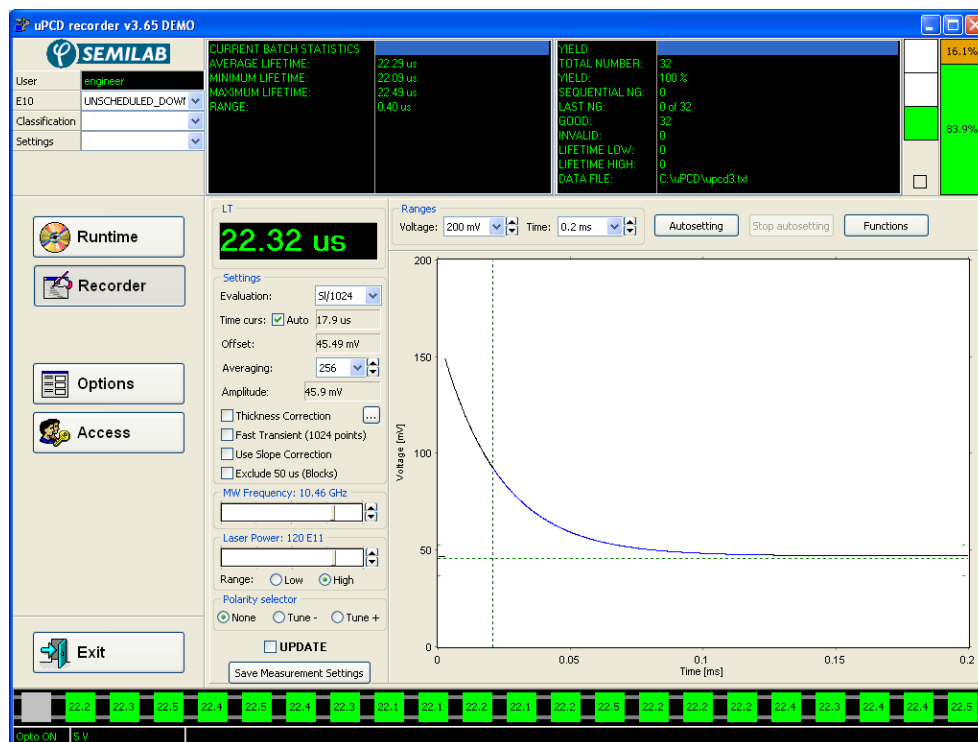


Figure 5-3: Recorder window

5.3.2.1 Autosetting button

Run the *Autosetting* process by clicking the **Autosetting** button at the top of the window. The *Microwave Frequency*, the *Averaging*, the signal *Range* (Y axis of the *Scope* window) and the *Time* window (X axis of the *Scope* window) are automatically set by the *Autosetting* process as described in section 3.6. The process can be halted by clicking on the **Stop Autosetting** button.

5.3.2.2 LT

The most recently measured lifetime value is displayed in the top left corner of the area. The color of the number depends on the Lifetime setup (see section 5.3.3.1).

5.3.2.3 Settings

Offset

The offset is the position of the horizontal dashed line in the *Scope* window, which is set by the Autosetting procedure so that the transient fits in the Scope window with the highest possible amplitude.

Averaging

Averaging is the number of transients taken and averaged for one measured value. The possible values are 2^n , where $0 \leq n \leq 12$, and whole number. More averaging means an increased signal to noise ratio at the expense of longer time necessary for the measurement.

Amplitude

The signal amplitude in the transient at the Time Cursor position is displayed also in the Settings panel in mV units. It may happen when parameters are set manually that the beginning of the transient is out of the Voltage range thus the displayed Amplitude value is not of the whole transient. One possibility is to drag the horizontal dashed line (**Offset**) and set to a position where the whole transient is inside the range. If the transient amplitude is higher than one third of the Voltage range it is recommended to set to higher range.

5.3.2.4 Pulse Width

The length of the laser pulse is adjustable between 266 ns and the 1/10 of the Time window. It is set automatically by the autosetting process. It is recommended not to reduce the pulse length manually to reach the highest signal-to-noise ratio.

5.3.2.5 Laser Power

The excitation level (number of excess carriers generated) is proportional to the laser power. By the application of different laser powers, transients at high and low excitation levels can be measured and compared leading to valuable information on the contamination species or oxide quality. In the default case high power is suggested (100%), this way a good signal to noise ratio is achieved.

5.3.2.6 Update

If this checkbox is selected, the measurement is continuously running, the lifetime value is refreshed.

5.3.2.7 Save Settings

All the above described settings can be saved to a recipe by this button. Recipe files have the extension '.rcp'. Recipes can be loaded in the general information area.

5.3.2.8 Ranges

Voltage

The Y-axis of the Scope window is the signal amplitude: the output voltage of the microwave detector. Depending on the signal level, different sensitivities can be selected such as 2.5, 5, 10, 20, 50, 100, 200, 500 mV and 1 and 2 V. In case of highly conductive material (0.1-1 Ωcm) the relative change in conductivity due to a given laser pulse is small. As a consequence high sensitivity should be selected. On the other hand, measurements on samples with high resistivity are normally performed with low sensitivity (large amplitude range) due to high signal amplitude. The proper selection of the sensitivity range is when the signal amplitude is between the 50% and 70% of the whole range. Use of *Autosetting* procedure is recommended.

Time

The X axis of the scope is the time elapsed after starting of the transient recording (the zero point is before excitation). The time ranges available are 0.008, 0.016, 0.04, 0.08, 0.2, 0.4, 1, 2, 4, 10, 20, 40 and 80 ms. In case of a proper setting the fitted decay time should be about 1/5 - 1/10 part of the whole range.

5.3.2.9 Scope window

The plot of the measured signal is shown here. The evaluated time constant of this transient is evaluated as lifetime.

5.3.2.10 Functions

Functions button contains the measuring possibility of the following functions:

Lifetime – Time

The lifetime as a function of the time curve is recorded in this window. The measuring interval can be set between 0-250 [sec] by dragging the dT slider. When dT is set to 0, the maximum available measuring speed is selected. Sampling determines how many measured values are averaged before recording a point on the plot.

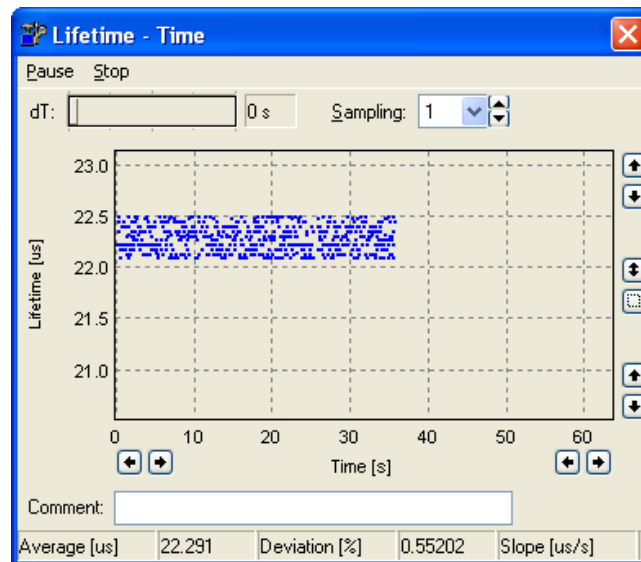


Figure 5-4: Lifetime – Time

Lifetime – Laser Power

The lifetime versus Laser Power curve is measured in this window. The resulting curve (logarithm of lifetime versus Laser Power) is typically increasing and it can be fitted with straight line. The slope of the line may be characteristic of the dominant contaminant in the wafer.

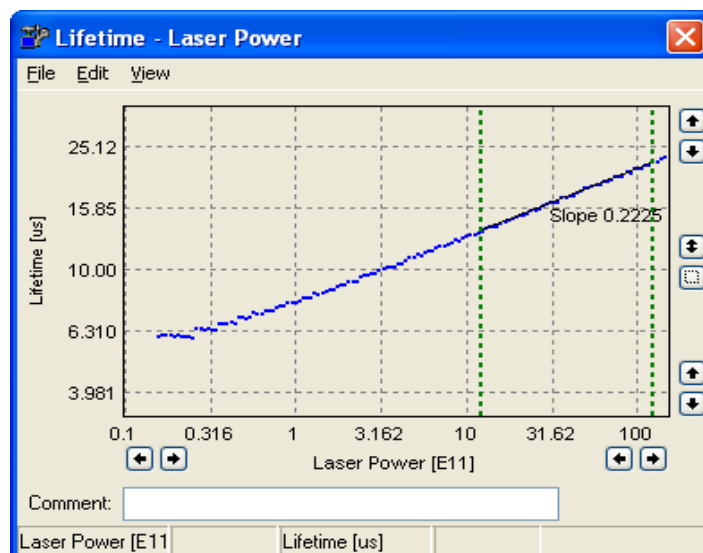


Figure 5-5: Lifetime – Laser Power

5.3.3 Options

Options window can be accessed only by an engineer level user.

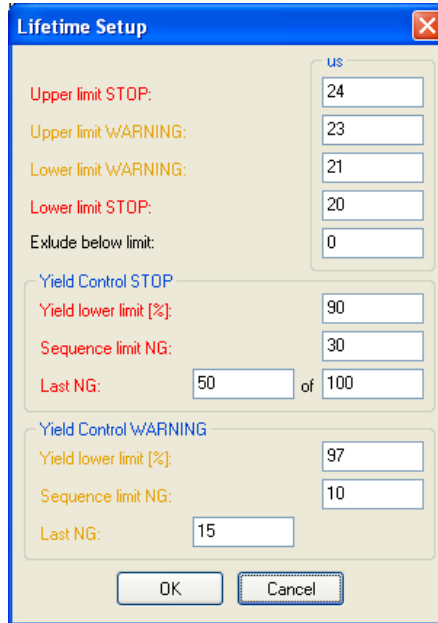
Setup	General settings	Optical Sensor
Lifetime Setup	☐ Update with Statistics ☐ Ignore Invalid	☒ Used Comparison Level Sense Delay
Save Classification	24V I/O Interface ☒ Not Used ☐ WinRT1 0x310 ☐ WinRT2 0x318 ☐ WinRT3 0x314	Serial TRG/WID ☐ Used Settings

Figure 5-6: Options window

5.3.3.1 Setup

Lifetime Setup

The upper and lower WARNING and STOP levels can be set in the appearing input window. It is also possible to set STOP and WARNING limits for the yield of the production.



Parameter	Value
Upper limit STOP:	24
Upper limit WARNING:	23
Lower limit WARNING:	21
Lower limit STOP:	20
Exclude below limit:	0
Yield Control STOP	
Yield lower limit [%]:	90
Sequence limit NG:	30
Last NG:	50 of 100
Yield Control WARNING	
Yield lower limit [%]:	97
Sequence limit NG:	10
Last NG:	15

Figure 5-7: Lifetime Setup

Save Classification

The classification setup can be saved here. Classification setup files have the .clf' extension.

Classifications can be loaded in the general information area.

5.3.3.2 General settings

Update with Statistics

In case this option is activated: when the *Update* checkbox is selected in the recorder window, the continuously measured data is added to the statistics.

Ignore Invalid

If checked, invalid points are not counted in statistics.

5.3.3.3 24V I/O Interface

In the WT-1200I system *Not Used* must be checked.

5.3.3.4 Optical sensor

If used, a signal of the optical sensor initializes a measurement.

<i>Comparison level</i>	If the signal of the optical sensor is higher than the comparison level, the sensor is on, and a sample is supposed to be under the head
<i>Sense Delay</i>	After the optical sensor is turned on, the system waits for the specified amount of time before starting the measurement. <i>Sense Delay</i> can be entered in units of ms.

5.3.4 Access levels

In the *Access* window, users can log in and off, and user administration can be managed. Two different access levels are defined in the μ -PCD operating software: Engineer level and Operator level.

- *Engineer*: This access level provides full functionality of the μ -PCD operating software.
- *Operator*: This access level has the following restrictions:
 - No access to the **Options** window including the I/O related settings, and classification settings. Classification can be changed only by loading classification files, previously made by an engineer.
 - No access to the measurement settings of the **Recorder**. Measurement settings can be changed only by loading recipes, previously made by an engineer.
 - **User Administration** is unavailable.

Administrators and engineers can create, delete and modify user accounts by clicking **User Administration**.

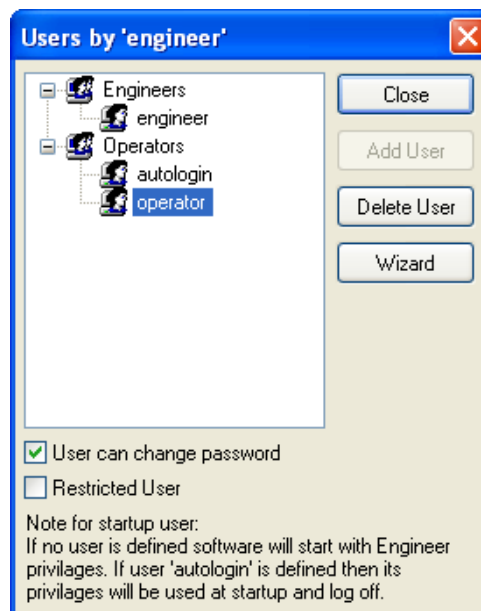


Figure 5-8: User administration window

! CAUTION: An Engineer level access user must be defined first. If only an operator access level is defined and the User Administration window is closed, the User Administration function becomes unavailable. This will lock the software into an operator access level. If only an operator level is created, a warning message is returned.

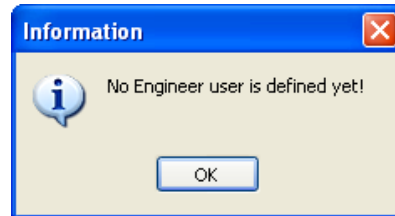


Figure 5-9: Information window

- After closing this warning message, do not close the *User Administration* window. Create an Engineer level user first.
- Users can be defined by clicking on the desired access level, then on **Add User**. The following window appears, prompting the user to define a user name and password.

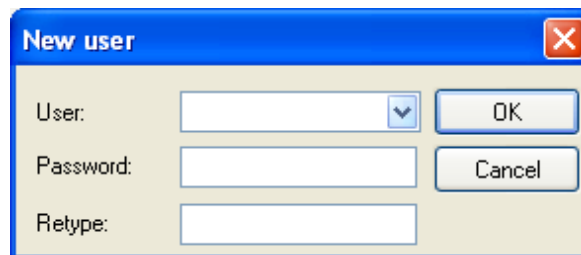


Figure 5-10: New user

✓ **NOTE:** If no password is defined, then only a user name needs to be entered at login.

- Users can be deleted with the **Delete User** button.
- An Engineer and Operator level user can automatically be created by clicking the **Wizard** button. The operator level user will be <autologin>. This option deletes all other defined users.
- After defining users, the μ -PCD operating software will start with a login window prompting the user to enter a user name and a password. If the login information is incorrect, the software cannot start.
- A default access level can be created for the μ -PCD software. This is done by creating a user with the name <autologin>. The software will then automatically start with the access level defined for the *autologin* user and will not ask for a password.
- A user change can then be performed via **Access/Log in as Different User**.
- Logging out is performed via **Access/Log Off**.
- A user's password can be changed from **Access/Change Password** if allowed.

5.4 General Information

General information is in the top left corner of the measurement window.

<i>User</i>	Under the Semilab logo, the currently logged in user name is visible.
<i>E10</i>	Status report according to the Semi10 standard.
<i>Classification</i>	The classification file of STOP and WARNING limits can be loaded here.
<i>Settings</i>	Recipes of measurement settings can be loaded here.

5.5 Statistics Area

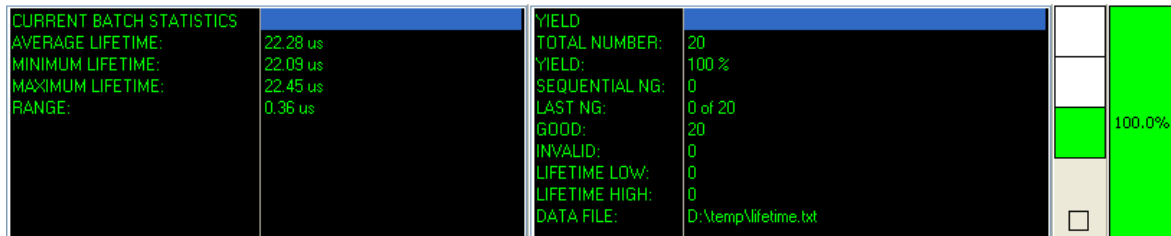


Figure 5-11: Statistics

The *Statistics* tab contains the following information fields: *Current Batch Statistics*, *Yield*.

Table 5-1: Current Batch Statistics

<i>Average lifetime</i>	The average lifetime of the current batch of measurements.
<i>Minimum lifetime</i>	The minimum measured value.
<i>Maximum lifetime</i>	The maximum measured value.
<i>Range</i>	The difference between minimum and maximum lifetime.

Table 5-2: Yield

<i>Total number</i>	Returns the total number of the measured samples.
<i>Yield</i>	Returns the ratio of the good measurements.
<i>Sequential NG</i>	Number of the samples, that sequentially returned lifetime values out of the stop limits.
<i>Last NG</i>	Specifies how many measurements were outside the stop limits when the last inadequate sample was measured.
<i>Good</i>	The number of wafers that had lifetime value within the stop limits.
<i>Invalid</i>	Returns number of wafers the tester was unable to measure.
<i>Lifetime low</i>	Number of the wafers that had a lifetime value below the lower stop limit.
<i>Lifetime high</i>	Number of the wafers that had a lifetime value above the upper stop limit.
<i>Data file</i>	Shows the path and the name of the file where all the statistics of lifetime and wafer numbers within the batch are saved in ASCII format (tab-delimited). All data in the subtab can be cleared with the Clear Statistics button. The software will then ask for the file name to save next statistical data to.

5.6 Measured value Indicator

At the bottom of the μ -PCD software window, there is a bar that indicates the results of the last several measurements.



The new measurement result (lifetime in μ s) shows up in the grey box at the left and, the box is colored according to the classification of the measurement. All the boxes are shifted right, and a new gray box appears for the next measurement.

5.7 Macros

By right-clicking in the mode selection area, a menu appears for macros. Besides the permanent menu items, the currently available macros are listed, and can be run from this list. A macro is available if it is in the directory of the software (c:\uPCD\) or in a subdirectory.

The Macro language has been implemented in μ -PCD to give users a wider range of control over software capabilities. It is a special program language that is mostly similar to BASIC or DOS batch programming.

5.7.1 Macro

Clicking *Macro/Macro* menu item the Macro Editor window appears:

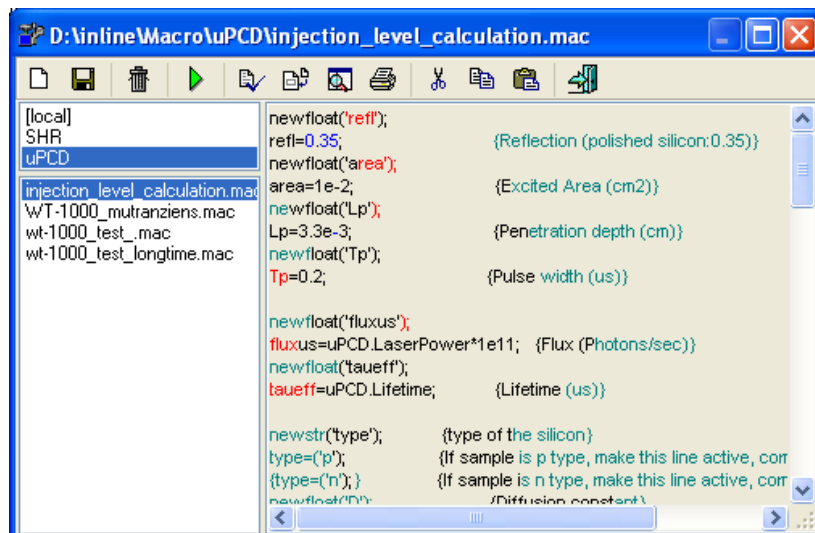


Figure 5-12: Macro window

The following tools are available for managing macros:

-  Creates a new macro.
-  Saves the current macro.
-  Deletes the selected macro.
-  Runs the selected macro.
-  Checks for Syntax Errors in the selected macro.

- | | |
|---|---|
|  | Lists all available macro variables, functions and procedures. |
|  | Opens the <i>Macro Watch</i> window for inspection of running macros. |
|  | Prints the selected macro. |
|  | Cuts the selected text from the selected macro. |
|  | Copies the selected text from the selected macro to the clipboard. |
|  | Pastes text from the clipboard to the selected macro. |
|  | Exits the macro window. |

5.7.1.1 Macro Syntax

For more detailed information on macro possibilities refer to the *Macro Syntax window* by pressing the  button in the *Macro Edit* window or *Macro/Macro Syntax* menu item.

5.7.1.2 Macro Watch

There is a debugger for macro language built in μ -PCD. You can add watches by *Macro/Watch* menu item. Right-click on the *Watch* window that opens to choose from the following options:

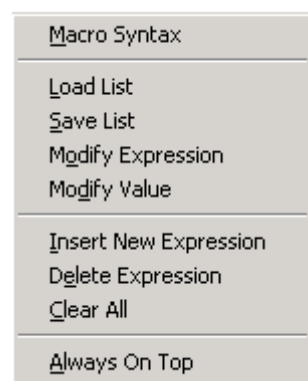


Figure 5-13: Macro watch menu

Clicking on *Macro Syntax* displays a list of all available variables, functions and procedures. Here, the user can check the syntax, find syntax errors and set the editor fonts properties.

6. Maintenance

The WT-1200I system has been designed to require very little periodic maintenance.

6.1 Cleaning the system

When using the WT-1200I in a cleanroom, cleaning may be required about every 6 months. When using the WT-1200I in common laboratory conditions, the cleaning procedure should be done every month. When using the WT-1200I in a factory environment, cleaning is required once a week.

Cleaning (decontamination) procedures should be performed with alcohol, using a cleanroom-compatible lint-free cloth.



Figure 6-1: Laser control box (the red arrows indicate the positions of the screws, the green arrow indicates the area to be cleaned with compressed air)

Clean the sensor, the fan filters and the laser control box with compressed air.

Dust can be removed from the not easily accessible parts of the WT-1200I by compressed air flow. If the top cover of the WT-1200I is removed, all parts can be cleaned by compressed air. The top cover can be removed with a 3 mm torque key. After cleaning, make sure to reconnect grounding of the top cover.

! CAUTION: Before removing the top cover, always make sure that the equipment is switched off.

7. Contacting Semilab

Only personnel trained and certified in maintenance of the WT-1200I may service the device. If you require help with the WT-1200I, contact Semilab for assistance. Semilab has world-wide service and support centers. Contact one of the locations listed below:

Semilab Headquarters

Office: Semilab Semiconductor Physics Laboratory Co. Ltd.

Address: Prielle Kornélia u. 2. H-1117 Budapest, Hungary

Tel: +36 1 505 4690

Fax: +36 1 505 4695

General information: semilab@semilab.com

Sales inquiries: sales.support@semilab.com

Technical inquiries: technical.support@semilab.com

South Europe: +33 6653 12642 | maximilien.dallery@semilab.com

Germany: +49 1742 433130 | michael.schweiger@semilab.com

China

Office: Semilab China

Address:

3rd Floor, B2 Building, BETWIN, No. 889, Shangcheng Road,
Pudong New Area, Shanghai, China

Tel: +86 21 58362889

Fax: +86 21 58362887

Email: info.china@semilab.com

Web: www.semilab.com.cn

Taiwan

Office: Semilab Taiwan Co. Ltd.

Address: 5F-5, No. 32, Gaotie 2nd Rd, Chubei City, Hsinchu
County, 30274, Taiwan.

Tel: +886 3 6672020

Fax: + 886 3 6672097

Email:

Web: www.semilab.com.tw

Japan

Office: Semilab Japan K.K.

Address: YS Shin-Yokohama Bldg. 6F, 2-15-10, Shin-Yokohama,
Kohoku-ku, Yokohama-shi Kanagawa 2220033 Japan

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#08-08 569880 Singapore
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Address: 101 Billerica Avenue, Building 5, Suite 105, North Billerica,
MA 01862
Tel: +1 508 647 8420
Fax: +1 508 647 8468
Email: info.usa@semilab.com

If you encounter a problem that appears to be a deficiency of the equipment, take detailed notes of the conditions leading to the problem, then consult Semilab for help. Semilab may suggest running tests that appear relevant to your problem. Document test results and error codes using the **Print Screen** function as described in the **Note** below.

✓ **NOTE:** Use the **Print Screen** function to capture the screens showing test results and error codes. When the appropriate screen appears, press the <Print Screen> button on your keyboard. Press <Ctrl-Esc> to display the Windows Start Menu. Select **Programs|Accessories|Paint** from the Start Menu to start the Paint program, Select **Edit|Paste** to paste the bitmap of the screen into the Paint program. Select **File|Print** to print the image or **File|Save** to save it to a file.